Chapter 3. Probabilistic Regression, Regularization and Kernel Ridge

Linear regression from a probabilistic view, regularization, ridge regression, the kernel trick

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1 Gaussian Likelihood Explains Squared Error

Theorem 3.1: MLE under Gaussian noise is least squares

If $y_i = \mathbf{x}_i^\top \boldsymbol{\beta} + \varepsilon_i$ with $\varepsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2)$, then

$$\log p(\mathbf{y} \mid X, \boldsymbol{\beta}, \sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\|\mathbf{y} - X\boldsymbol{\beta}\|^2.$$

For fixed σ^2, maximizing the likelihood is minimizing the SSE, so $\hat{\boldsymbol{\beta}}_{\text{MLE}}$ solves the normal equations. Maximizing over σ^2 then gives $\hat{\sigma}^2 = \text{SSE}/n$.

Proof. The density of y_i is $(2\pi\sigma^2)^{-1/2}\exp\{-(y_i - \mathbf{x}_i^\top\boldsymbol{\beta})^2/(2\sigma^2)\}$; independence multiplies densities and the log turns the product into the displayed sum. Differentiating in σ^2: $-\frac{n}{2\sigma^2} + \frac{\text{SSE}}{2\sigma^4} = 0$. ■

2 Ridge Regression

2.1 Definition and closed form

Definition 3.2: ridge

For $\lambda > 0$, $J(\boldsymbol{\beta}) = \frac{1}{2}\|\mathbf{y} - X\boldsymbol{\beta}\|^2 + \frac{\lambda}{2}\|\boldsymbol{\beta}\|^2$.

Theorem 3.3: ridge solution

$\nabla J = X^\top X\boldsymbol{\beta} - X^\top\mathbf{y} + \lambda\boldsymbol{\beta}$ and the unique minimizer is $\hat{\boldsymbol{\beta}}_\lambda = (X^\top X + \lambda\mathbf{I})^{-1}X^\top\mathbf{y}$, even when X is rank deficient.

Proof. $\mathbf{v}^\top(X^\top X + \lambda\mathbf{I})\mathbf{v} = \|X\mathbf{v}\|^2 + \lambda\|\mathbf{v}\|^2 > 0$ for $\mathbf{v} \neq \mathbf{0}$, so the matrix is positive definite, invertible, and J is strictly convex. ■

2.2 MAP interpretation

With prior $\boldsymbol{\beta} \sim N(\mathbf{0}, \tau^2\mathbf{I})$, $-\log p(\boldsymbol{\beta} \mid \mathbf{y}) = \frac{1}{2\sigma^2}\|\mathbf{y} - X\boldsymbol{\beta}\|^2 + \frac{1}{2\tau^2}\|\boldsymbol{\beta}\|^2 + \text{const}$; multiplying by σ^2 gives ridge with $\lambda = \sigma^2/\tau^2$. A tight prior (small τ) means strong shrinkage.

2.3 Shrinkage in singular directions

Let $X = U\Sigma V^\top$ (thin SVD, singular values s_i). Since $X^\top X + \lambda\mathbf{I} = V(\Sigma^2 + \lambda\mathbf{I})V^\top$,

$$\hat{\boldsymbol{\beta}}_\lambda = \sum_i \frac{s_i}{s_i^2 + \lambda}(\mathbf{u}_i^\top\mathbf{y})\mathbf{v}_i, \qquad X\hat{\boldsymbol{\beta}}_\lambda = \sum_i \frac{s_i^2}{s_i^2 + \lambda}(\mathbf{u}_i^\top\mathbf{y})\mathbf{u}_i.$$

Least squares fits $\sum_i(\mathbf{u}_i^\top\mathbf{y})\mathbf{u}_i$; ridge multiplies the ith component by $s_i^2/(s_i^2 + \lambda) \in (0, 1)$. Small singular values (directions along which the data barely vary, where least squares amplifies noise by $1/s_i$) are shrunk most. $\lambda \to 0$ recovers least squares; $\lambda \to \infty$ sends coefficients to zero.

2.4 Bias and variance of ridge

Assume $\mathbf{y} = X\boldsymbol{\beta}_0 + \boldsymbol{\varepsilon}$ with $\mathbb{E}\boldsymbol{\varepsilon} = \mathbf{0}$, $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}$, and X of full column rank with $X^\top X = VDV^\top$, $D = \text{diag}(d_1, \ldots, d_p)$, $d_i > 0$.

Theorem 3.4: ridge trades bias for variance

(i) $\mathbb{E}\hat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta}_0 = -\lambda(X^\top X + \lambda \mathbf{I})^{-1}\boldsymbol{\beta}_0$; for $\lambda > 0$ the estimator is unbiased iff $\boldsymbol{\beta}_0 = \mathbf{0}$.

(ii) $\operatorname{Cov}(\hat{\boldsymbol{\beta}}_\lambda) = \sigma^2(X^\top X + \lambda \mathbf{I})^{-1} X^\top X (X^\top X + \lambda \mathbf{I})^{-1}$ and

$$\operatorname{tr}\operatorname{Cov}(\hat{\boldsymbol{\beta}}_\lambda) = \sigma^2 \sum_i \frac{d_i}{(d_i+\lambda)^2} < \sigma^2 \sum_i \frac{1}{d_i} = \operatorname{tr}\operatorname{Cov}(\hat{\boldsymbol{\beta}}_0) \qquad (\lambda > 0).$$

(iii) $\lambda \mapsto \|\hat{\boldsymbol{\beta}}_\lambda\|$ is decreasing.

Proof. (i) $\mathbb{E}\hat{\boldsymbol{\beta}}_\lambda = (X^\top X + \lambda \mathbf{I})^{-1} X^\top X \boldsymbol{\beta}_0$. From $(X^\top X + \lambda \mathbf{I})^{-1}(X^\top X + \lambda \mathbf{I}) = \mathbf{I}$ we get $(X^\top X + \lambda \mathbf{I})^{-1} X^\top X = \mathbf{I} - \lambda(X^\top X + \lambda \mathbf{I})^{-1}$, which gives the bias. The matrix $(X^\top X + \lambda \mathbf{I})^{-1}$ is invertible, so the bias vanishes iff $\boldsymbol{\beta}_0 = \mathbf{0}$.

(ii) $\hat{\boldsymbol{\beta}}_\lambda = M\mathbf{y}$ with $M = (X^\top X + \lambda \mathbf{I})^{-1} X^\top$, so $\operatorname{Cov} = M(\sigma^2 \mathbf{I})M^\top$. All three matrices are diagonalized by V, giving eigenvalues $\sigma^2 d_i/(d_i+\lambda)^2$; the trace is their sum, and $(d_i+\lambda)^2 > d_i^2$.

(iii) $\|\hat{\boldsymbol{\beta}}_\lambda\|^2 = \sum_i s_i^2 (\mathbf{u}_i^\top \mathbf{y})^2/(s_i^2+\lambda)^2$ by orthonormality of $\mathbf{v}_i$; each term decreases in λ. ■

3 Other Penalties and the Lasso

More generally one penalizes $\sum_j |\beta_j|^q$. $q = 2$ is ridge (round level sets, pure shrinkage); $q = 1$ is the lasso, whose diamond-shaped level sets have corners on the axes and produce exact zeros; $0 < q < 1$ is nonconvex.

Theorem 3.5: soft thresholding

$\arg\min_b \ \frac{1}{2}(b-z)^2 + \lambda|b| = \operatorname{sign}(z)\max(|z|-\lambda, 0)$, whereas $\arg\min_b \frac{1}{2}(b-z)^2 + \frac{\lambda}{2}b^2 = z/(1+\lambda)$.

Proof. The objective is strictly convex; the optimality condition is $0 \in b - z + \lambda\,\partial|b|$, where $\partial|b| = \{\operatorname{sign} b\}$ for $b \neq 0$ and $[-1, 1]$ at 0. If $b > 0$: $b = z - \lambda$, valid iff $z > \lambda$. If $b < 0$: $b = z + \lambda$, valid iff $z < -\lambda$. If $b = 0$: need $z \in \lambda[-1, 1]$, i.e. $|z| \leq \lambda$. The ridge case is the stationarity condition $b - z + \lambda b = 0$. ■

Lasso moves every coefficient by a fixed amount and cuts small ones to zero (sparsity); ridge rescales proportionally and never produces exact zeros.

4 Kernel Ridge Regression

4.1 Feature-space and sample-space forms

Let $\boldsymbol{\phi}(\mathbf{x}) \in \mathbb{R}^m$ be a feature map and, $\Phi = [\boldsymbol{\phi}(\mathbf{x}_1) \cdots \boldsymbol{\phi}(\mathbf{x}_n)] \in \mathbb{R}^{m \times n}$ (columns are samples). Ridge in feature space minimizes $J = \frac{1}{2}\|\mathbf{y} - \Phi^\top \mathbf{w}\|^2 + \frac{\lambda}{2}\|\mathbf{w}\|^2$, so

$$\nabla J = -\Phi\mathbf{y} + (\lambda \mathbf{I}_m + \Phi\Phi^\top)\mathbf{w}, \qquad \hat{\mathbf{w}} = (\lambda \mathbf{I}_m + \Phi\Phi^\top)^{-1}\Phi\mathbf{y}.$$

Theorem 3.6: push-through identity

$(\lambda \mathbf{I}_m + \Phi\Phi^\top)^{-1}\Phi = \Phi(\lambda \mathbf{I}_n + \Phi^\top\Phi)^{-1}$. Hence with the Gram matrix $K = \Phi^\top\Phi$, $K_{ij} = k(\mathbf{x}_i, \mathbf{x}_j) = \boldsymbol{\phi}(\mathbf{x}_i)^\top \boldsymbol{\phi}(\mathbf{x}_j)$,

$$\hat{\mathbf{w}} = \Phi\boldsymbol{\alpha}, \quad \boldsymbol{\alpha} = (K + \lambda \mathbf{I}_n)^{-1}\mathbf{y}, \qquad f(\mathbf{x}) = \boldsymbol{\phi}(\mathbf{x})^\top \hat{\mathbf{w}} = \sum_{i=1}^{n} \alpha_i k(\mathbf{x}_i, \mathbf{x}) = \mathbf{k}_{\mathbf{x}}^\top (K + \lambda \mathbf{I}_n)^{-1}\mathbf{y}.$$

Proof. $\Phi(\lambda \mathbf{I}_n + \Phi^\top\Phi) = \lambda\Phi + \Phi\Phi^\top\Phi = (\lambda \mathbf{I}_m + \Phi\Phi^\top)\Phi$. Multiply on the left by $(\lambda \mathbf{I}_m + \Phi\Phi^\top)^{-1}$ and on the right by $(\lambda \mathbf{I}_n + \Phi^\top\Phi)^{-1}$ (both positive definite). The two identity matrices have different sizes. ■

Remark 3.7: the kernel trick and cost

$\boldsymbol{\phi}$ appears only through inner products, so only the kernel k is needed. The primal form solves an $m \times m$ system and the dual an $n \times n$ one: the kernel form wins when $m \gg n$ (e.g. the Gaussian kernel, whose feature space is infinite-dimensional). Merely mapping to higher dimensions is feature engineering; computing through k without forming $\boldsymbol{\phi}$ is the kernel trick. An alternative derivation substitutes $\mathbf{w} = \Phi\boldsymbol{\alpha}$ and obtains $K[(K + \lambda \mathbf{I})\boldsymbol{\alpha} - \mathbf{y}] = \mathbf{0}$; when K is singular one may not cancel K, but the displayed $\boldsymbol{\alpha}$ is a solution and all solutions give the same $\hat{\mathbf{w}}$.

4.2 Kernels

- Linear: $k(\mathbf{x}, \mathbf{z}) = \mathbf{x}^\top \mathbf{z}$.
- Quadratic in 2D: $(\mathbf{x}^\top \mathbf{z})^2 = x_1^2 z_1^2 + x_2^2 z_2^2 + 2x_1 x_2 z_1 z_2$, so $\boldsymbol{\phi}(\mathbf{x}) = (x_1^2, x_2^2, \sqrt{2}x_1 x_2)$. In 1D, $(1 + xz)^2$ has $\boldsymbol{\phi}(x) = (1, \sqrt{2}x, x^2)$.
- Gaussian: $k(\mathbf{x}, \mathbf{z}) = \exp(-\|\mathbf{x} - \mathbf{z}\|^2/(2\sigma^2))$; σ is the bandwidth.
- Normalized: $\tilde{k}(\mathbf{x}, \mathbf{z}) = k(\mathbf{x}, \mathbf{z})/\sqrt{k(\mathbf{x}, \mathbf{x})k(\mathbf{z}, \mathbf{z})}$.

Theorem 3.8: Gram matrices are PSD

For any feature-map kernel and any $\mathbf{a} \in \mathbb{R}^n$, $\mathbf{a}^\top K\mathbf{a} = \|\sum_i a_i \boldsymbol{\phi}(\mathbf{x}_i)\|^2 \geq 0$.

Not every similarity function is a valid kernel; positive semidefiniteness of all Gram matrices is the requirement.

4.3 Worked examples

Example 3.9: linear kernel

$\mathbf{x}_1 = (1, 2)$, $\mathbf{x}_2 = (2, 0)$, $\mathbf{y} = (3, 1)$, $\lambda = 1$: $K = \begin{pmatrix} 5 & 2 \\ 2 & 4 \end{pmatrix}$, $(K + \mathbf{I})^{-1} = \frac{1}{26}\begin{pmatrix} 5 & -2 \\ -2 & 6 \end{pmatrix}$, $\boldsymbol{\alpha} = (1/2, 0)$, $\hat{\mathbf{w}} = \frac{1}{2}\mathbf{x}_1 = (1/2, 1)$. At $\mathbf{x} = (1, 0)$: $\mathbf{k}_{\mathbf{x}} = (1, 2)$ and $f(\mathbf{x}) = 1/2 = \hat{\mathbf{w}}^\top \mathbf{x}$.

Example 3.10: quadratic kernel

$\mathbf{x}_1 = (1,1)$, $\mathbf{x}_2 = (2,0)$, $\mathbf{y} = (3,2)$, $k = (\mathbf{x}^\top \mathbf{z})^2$, $\lambda = 1$. Then $K_{11} = 4$, $K_{12} = 4$, $K_{22} = 16$,

$$(K + \mathbf{I})^{-1} = \frac{1}{69}\begin{pmatrix} 17 & -4 \\ -4 & 5 \end{pmatrix}, \qquad \boldsymbol{\alpha} = \frac{1}{69}\begin{pmatrix} 43 \\ -2 \end{pmatrix}.$$

At $\mathbf{x}_t = (1,0)$, $\mathbf{k}_{\mathbf{x}_t} = (1,4)$ and $f(\mathbf{x}_t) = (43-8)/69 = 35/69 \approx 0.507$. Check in feature space: $\boldsymbol{\phi}(\mathbf{x}_1) = (1,1,\sqrt{2})$, $\boldsymbol{\phi}(\mathbf{x}_2) = (4,0,0)$, $\hat{\mathbf{w}} = \Phi\boldsymbol{\alpha} = (35, 43, 43\sqrt{2})/69$ and $\boldsymbol{\phi}(\mathbf{x}_t) = (1,0,0)$ give the same $35/69$.